

Constraint, Coordinate, and Flow: An Operational Anatomy of the Christoffel Rank Read

the Capacity Clock, the Affine Sort, Isospectral Motion, and the Execution Fingerprint Below the Floor

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Abstract

This paper reports an operational anatomy of the Christoffel Rank Read: a sorting of every quantity the instrument produces into the layer it actually belongs to, obtained by measurement rather than by stipulation. The instrument is the LDL^T (equivalently Cholesky) factorization of the Hankel moment matrix of a finite positive measure, together with the three-term recurrence of its orthogonal polynomials. Four experiment sets, each preregistered, were run against it.

First, a clock law. The arithmetic cost of the factorization is $c = 2\log_{10}(r_{\text{out}}/\text{cap})$ digits of cancellation per resolved pivot — a conformal invariant of the support read jointly with the origin of coordinates — verified within ten percent on four node families. The residual five-percent deficit, previously flagged as an unexplained systematic, is here identified as the finite-index Szegő prefactor: on the equilibrium measure a three-parameter fit recovers the leading capacity constant to one part in a thousand with a logarithmic

exponent $\gamma = -0.47$, matching the arcsine soft-edge value $-1/2$; on non-equilibrium measures the prefactor is subdominant to the constrained-equilibrium acceleration, which is the same finite-index physics seen from the other side.

Second, the instrument wall. The precision-limited false wall is a predictable function of the budget and the object's cancellation profile, $W_{\text{instr}}(\text{dps}) = D^{-1}(\text{dps} - \kappa) + G$ with headroom $\kappa = 0 \pm 1$; calibrated on one family it predicted false walls blind on two others within a preregistered two-step band, and in every starved run the wall was announced by the first negative sub-floor pivot — a sign flip — recovering the detector law of prior work from an independent direction.

Third, an affine sort. Under translation, dilation, and reflection of the read-frame, the object wall is exactly invariant, the Jacobi off-diagonal coefficients are frame-blind, the diagonal coefficients carry the frame and nothing else, and the clock covaries exactly as the capacity law dictates. The sort refutes, inside this system, the strong relational thesis that no coordinate-free quantities exist, while confirming that every continuous instrument-side quantity is a ratio, a difference, or a sign. What survives every admissible frame is an integer: the count of distinctions.

Fourth, two motions. The read-frame group (affine maps of the support) fixes the measure and moves the coordinates. The isospectral flow group (Toda) fixes the spectrum and moves the measure: under it the wall and the leading clock are exactly conserved while the representation flows and the finite-index prefactor drifts, tying the execution layer to the subleading correction. Finally, a discriminator built to answer whether the "stream-first" reading of these results is testable: two numerically distinct routes to the same pivot sequence agree to 10^{-378} above the floor and decorrelate to chance below it, with the sub-floor sign shown to be a deterministic fingerprint of the algorithm-precision pair and independent of the object. The rivalry between "reality is states" and "reality is a stream" dissolves into a measured tier assignment: the object is state-first and exact; the instrument's exhaust is stream-first and carries the process, not the data. The paper closes with the finite-to-infinite passage of the count invariant — where a count must become a counting function — as the gated Clay-adjacent experiment.

1. Anchor: what is being measured, and why

Two of the Clay Millennium problems are within arm's reach of this instrument, and naming them keeps the work from wandering. On the Birch–Swinnerton-Dyer side, both channels already reproduce from the local data a_p alone: rank certificates and leading coefficients, computed to ten digits in the program's Scale Notebook, with a rank-four vanishing certificate confirmed and a wall law verified into the L-function regime. On the Riemann side, the pending experiment feeds prime-derived moments at growing budgets and asks whether the wall tracks the zero-counting function $N(T)$. That run is not readable until one thing is built: a way to separate what the *instrument* says at a finite precision from what the *object* says, at every step. This paper builds that separation, and building it forces a larger accounting — because once you can say which part of a reading is the object and which is the apparatus, you are obliged to sort every quantity the program has ever produced.

The prior installment established that the wall is not one wall but three, cleanly separable: an *object* wall that is algebraic and exact, an *instrument* wall set by the precision floor, and a *detector* wall governed by the sign

statistics of the sub-floor residue. That separation is the spine of everything here. The present paper makes the instrument wall quantitative and predictive (§4), gives the instrument a clock with a closed-form rate and a derived correction (§3), identifies the detector wall's announcement mechanism as a single sign bit (§4), and then adds two axes the earlier work did not have: the read-frame (§5) and the isospectral flow (§6). The synthesis (§9) sorts the accumulated results into an architecture whose apex is constraint — a viewpoint that arose across a sustained multi-model dialogue and which the measurements, not the eloquence, earn a place in the record.

A word on scope, stated once and meant throughout. The measurements below are rigorous *within this system*: a finite positive measure, its moments, its orthogonal polynomials, its factorization. They establish nothing about the physical universe. What they do is convert several intuitions that had lived only as metaphor — “the wall is an instrument floor,” “only differences survive,” “reality is a stream, not a sequence of states” — into properties that are either true or false of a concrete mathematical object, and then measure which. That conversion is the whole contribution. A metaphor that can be measured has become a hypothesis; a hypothesis that survives its own preregistered test has earned a provisional place; and the boundary between “productive lens” and “claim about the world” is marked at every step rather than blurred.

2. The classical instrument: moment problem, positivity, determinacy, recurrence

Before any measurement, the apparatus must be shown sound, and its soundness is entirely classical. The relevant theory is the moment problem, and stating it precisely also supplies the asymptotic machinery §3 will need.

The classical moment problem asks when a sequence of real numbers $(s_n)_{n \geq 0}$ is the moment sequence $s_n = \int x^n d\mu(x)$ of some positive measure μ on the real line. The question splits by support: the Hamburger problem on all of $\mathbb{R}$, the Stieltjes problem on the half-line $[0, \infty)$, and the Hausdorff problem on a bounded interval. The existence criterion for the Hamburger case is a positivity condition on the associated Hankel matrix H , with entries $H_{i,j} = s_{i+j}$. Necessity is immediate: for any real polynomial $P(x) = \sum_i c_i x^i$, the integral of its square against a positive measure cannot be negative, and expanding that integral yields exactly the quadratic form $c^T H_n c \geq 0$, so every truncated Hankel matrix is positive semidefinite. Sufficiency is Haviland's theorem, which leans on the Riesz representation theorem together with the fact that in one variable every non-negative polynomial is a sum of squares; a functional positive on sums of squares is therefore positive on all non-negative polynomials and extends to a representing measure. This is the ground the whole program stands on: **positivity of the Hankel matrix is existence of the object**. [LOCKED, classical]

If a representing measure exists, uniqueness is the next question — determinacy versus indeterminacy — and it is governed by growth. If the moments grow too quickly the solution set inflates into infinitely many distinct measures sharing one moment sequence. Carleman's condition gives a sufficient criterion for determinacy: divergence of $\sum_n s_{2n}^{-1/2n}$ forces the tail to decay fast enough to pin the measure. The same fact wears two other faces. In the recurrence-coefficient picture, divergence of $\sum b_n^{-1}$ over the off-diagonal Jacobi

coefficients is sufficient for determinacy. In the spectral picture, the problem is determinate exactly when the smallest eigenvalue of the truncated Hankel matrix H_n tends to zero as $n \rightarrow \infty$. That last equivalence is the bridge this program lives on: **the geometry of the moment space and the numerical conditioning of the Hankel system are the same phenomenon**. The smallest eigenvalue going to zero is, numerically, the statement that the matrix is becoming singular — that a wall is approaching — and it is, simultaneously, the analytic statement that the measure is determined. The wall is not a numerical nuisance sitting on top of clean mathematics; it *is* the mathematics, read through a finite instrument. [LOCKED, classical]

From a positive measure with finite moments one constructs monic orthogonal polynomials $P_n(x)$ by Gram–Schmidt on the monomials $\{1, x, x^2, \dots\}$ in $L^2(\mathbb{R}, d\mu)$. Their defining structural signature is the three-term recurrence: for the orthonormal polynomials p_n ,

$$x p_n(x) = a_{n+1} p_{n+1}(x) + b_{n+1} p_n(x) + a_n p_{n-1}(x),$$

with $a_n > 0$ and b_n real. The proof is pure orthogonality: $x p_n$ has degree $n + 1$, so it expands in the basis, and self-adjointness of multiplication by x kills every coefficient below $n - 1$, leaving three surviving terms. Encoding these coefficients as a tridiagonal matrix produces the Jacobi operator J , whose spectrum is the support of μ and whose principal-minor structure is exactly the Cholesky/ LDL^T sweep of H . Three descriptions — measure, moments, Jacobi operator — are one object in different coordinates, and the pivots of the factorization are the squared norms $h_n = \|P_n\|^2$ of the monic polynomials. Everything measured in this paper is a functional of these h_n and of the coefficients a_n, b_n ; the classical theory guarantees they exist, are unique for a determinate measure, and are computable by two different routes — a fact §7 turns into an experiment.

The program’s name points at a specific classical object these constructions make available. Writing $K_n(x, y) = \sum_{k=0}^n p_k(x) p_k(y)$ for the Christoffel–Darboux kernel of the orthonormal polynomials, the **Christoffel function** is $\lambda_n(x) = 1/K_n(x, x)$ — the solution of the extremal problem $\lambda_n(x_0) = \min\{\int |P|^2 d\mu : P(x_0) = 1, \deg P \leq n\}$, the least mass a degree- n polynomial can concentrate at x_0 while staying normalised there. Two facts make it the natural instrument. First, the Christoffel numbers $\lambda_n(x_i)$ at the zeros of p_{n+1} are exactly the Gauss quadrature weights, so the Christoffel function is the bridge between the continuous measure and its discrete quadrature image — the same bridge §3 exploits when it identifies unit weights on Chebyshev nodes with Gauss–Chebyshev weights. Second, by the Máté–Nevai–Totik theorem $n \lambda_n(x)$ converges almost everywhere in the bulk to the density of μ relative to the equilibrium measure of its support, so the Christoffel function reads, in the large- n limit, precisely the quantity that governs whether a family shifts gears or runs a single clock (§3). The “rank read” half of the name is the other side of the same coin: for a measure of exactly M atoms the Hankel matrix has rank M , its Cholesky sweep yields M positive pivots and a vanishing $(M + 1)$ -th, and the wall is the rank plus one. Reading the rank of the moment matrix is reading the number of atoms in the measure, and the Christoffel function is how that count is distributed across the support. The program’s name is a compression of exactly the object this paper anatomises. [LOCKED, classical]

The asymptotics that §3 requires are the Szegő theorems. For a measure on $[-1, 1]$ in the Szegő class, the norm ratios $a_n^2 = h_n/h_{n-1}$ converge to $(b - a/4)^2$, the square of the logarithmic capacity of the support; the pivots therefore decay geometrically at a rate fixed by capacity alone, with a subleading prefactor fixed by the *edge behaviour* of the density. A square-root (arcsine) singularity at the endpoints and a flat (hard-

edge) density give different prefactors — a distinction that will turn out to be measurable, and to explain the one loose thread in the clock law.

3. The capacity clock and its subleading correction

Let μ be an M -point positive measure with moments $\mu_k = \sum_i w_i x_i^k$ read from an origin at zero, and let H be its Hankel matrix. The i -th pivot of the Cholesky sweep is the monic norm h_i ; the largest mass entering that pivot's computation is of order μ_{2i} . Define the **cancellation depth**

$$D(i) = \log_{10} \frac{\text{big}_i}{|h_i|}, \quad \text{big}_i = \max(|\mu_{2i}|, \text{subtracted mass}),$$

the number of decimal digits the instrument must destroy to resolve pivot i . This reframes precision as a budget. Each resolved pivot is one tick of the factorization; $D(i)$ is the digits spent on tick i ; and the local slope $c = dD/di$ is the burn rate — the clock speed of this object read from this frame. The name is not decorative: §5 will show c behaves under coordinate change exactly as a clock rate should.

Potential theory fixes the rate. With $r_{\text{out}} = \max_i |x_i|$ the outer radius from the read origin and cap the logarithmic capacity of the support (for an interval $[a, b]$, $\text{cap} = (b - a)/4$), the Szegő asymptotics give $\mu_{2i} \sim r_{\text{out}}^{2i}$ and $h_i \sim \text{cap}^{2i}$, so

$$c = 2 \log_{10} \frac{r_{\text{out}}}{\text{cap}} \text{ digits per tick.}$$

The prediction was preregistered against four node families at $M = 30$, $\text{dps} = 400$, with the early-window slope taken over $i \in [3, 12]$:

family	wall	predicted c	early c	late c	gear change	P1 (10%)
random $(-10, 10)$	31	0.626	0.636	1.367	yes	PASS
integer $1..M$	31	1.234	1.191	1.605	yes	PASS
chebyshev $[-1, 1]$	31	0.602	0.571	0.592	no	PASS
equispaced $[-1, 1]$	31	0.602	0.571	0.979	yes	PASS

Four for four within the band. Two structural facts deserve isolating. First, **scale cancels completely**: the rate depends only on the ratio $r_{\text{out}}/\text{cap}$, so any interval centred on the read origin burns $2 \log_{10} 2 \approx 0.602$ digits per tick whether it spans $[-1, 1]$ or $[-10^{10}, 10^{10}]$. The clock reads the *shape* of the support and its position relative to the read-head; it is blind to size. In the program's older language, this is "shape before value" appearing as literal arithmetic. Second, the deficit: every early-window slope sits about five percent below prediction, and the offset is *shared* across all four families, which marks it as systematic rather than noise.

The rigorous reference resolves that deficit, and the resolution is the edge-behaviour prefactor from §2. For an arcsine (soft-edge) density the monic norm carries a logarithmic correction, giving

$$D(i) \sim 2 \log_{10}(2) i - 1/2 \log_{10} i + \text{const},$$

so a finite-window slope is depressed below the leading constant by roughly $1/2 \langle 1/i \rangle / \ln 10$, which over $[3,12]$ is about 0.035 — closing almost exactly the gap between the predicted 0.602 and the measured 0.571. To test this directly, fit the three-parameter model $D(i) = c i + \gamma \log_{10} i + b$ over $i \in [3,27]$ and read off both the corrected leading constant c and the logarithmic exponent γ :

family	edge class	leading c	raw slope $[3,12]$	raw error	fit c	fit error	fit γ
chebyshev	arcsine	0.602	0.571	5.1%	0.602	0.1%	−0.471
random $(-10,10)$	flat	0.626	0.636	1.6%	—	—	—
integer $1..M$	flat	1.234	1.191	3.5%	—	—	—
equispaced	flat	0.602	0.571	5.2%	—	—	—

On the equilibrium (Chebyshev) measure the correction is decisive: the fit recovers the leading capacity constant to **one part in a thousand** ($c = 0.602$) with $\gamma = -0.471$, matching the arcsine soft-edge value $-1/2$ to within finite-index scatter. **The five-percent deficit is the predicted Szegő prefactor, not an error.** [PULL — clean single-machine confirmation on the equilibrium measure]

The two edge classes can be derived explicitly, and the derivation is what licenses reading γ as a diagnostic. For the arcsine (Chebyshev) case the monic-Chebyshev norms are $h_n = \pi 2^{1-2n}$ for $n \geq 1$ and the continuum arcsine moments are $\mu_{2n} = \binom{2n}{n} 4^{-n}$, so with $\binom{2n}{n} \sim 4^n / \sqrt{\pi n}$,

$$D(n) = \log_{10} \mu_{2n} - \log_{10} h_n = \log_{10} \binom{2n}{n} - \log_{10} (2\pi) \\ \sim 2 \log_{10} (2) n - 1/2 \log_{10} n - 1/2 \log_{10} \pi - \log_{10} (2\pi).$$

The leading slope is $2 \log_{10} 2 = 0.602$ and the coefficient of $\log_{10} n$ is $-1/2$, the value the fit returned as -0.471 . For the flat (Legendre) case the monic norms satisfy $h_n \sim (\pi/2) 4^{-n}$ — the polynomial prefactor cancels — while the uniform moments are $\mu_{2n} = 1/(2n+1)$, giving $D(n) \sim 2 \log_{10} (2) n - \log_{10} (2n+1) + \text{const}$, so the coefficient of $\log_{10} n$ is -1 , twice the arcsine value. The two endpoint singularities are therefore distinguishable by a single measured exponent — in principle a spectroscopy of the edge behaviour from the cancellation profile alone. In practice this separation is isolable only on the equilibrium measure, because on the non-equilibrium families the constrained-equilibrium acceleration adds its own curvature: over a short window the linear and logarithmic terms are collinear and the fit cannot separate them, and over a long window the changing slope violates the model. The clean spectroscopy and the gear change are the same finite-index physics, and only one can be isolated at a time.

For the non-equilibrium families the three-parameter fit does not stabilise, and the reason is not a failure of the correction but the presence of a larger effect. These families accelerate — their late-window slope exceeds their early-window slope, the “gear change” recorded in prior work — so a single linear-plus-logarithmic model over a wide window is misspecified: the slope itself is changing. The clean subleading correction therefore lives exactly where the clean single clock lives, on the equilibrium measure, and everywhere else it is dominated by the constrained-equilibrium acceleration. That the correction and the acceleration cannot be disentangled by a global fit is itself the consistent statement, because they are two views of one finite-index phenomenon: the departure of the discrete measure from the interval’s equilibrium density.

That acceleration has a theorem behind it. The Chebyshev family is not merely the best-behaved row of the table; unit weights on Chebyshev nodes are proportional to the Gauss–Chebyshev quadrature weights, which are all equal, and Gauss quadrature on M nodes is exact through degree $2M - 1$. The discrete moments therefore equal a constant times the continuum arcsine moments *exactly* through μ_{2M-1} , and the discrete pivots are the continuum pivots all the way to the wall. Measured: a slope of 0.586 over $i \in [3,29]$ with a maximum deviation from a straight line of 0.106 digits — a single clock, no gears, because Chebyshev nodes sit on the equilibrium measure of the interval and there is no discreteness constraint left to bind. [LOCKED — Gauss exactness is classical; the residual is measured] Every family that departs from that equilibrium density pays a growing surcharge, the Runge-side dive, and for the integer family the local slope climbs from 1.19 to 2.24 across the sweep, with a twenty-five-percent-departure threshold landing near $t = i/M = 0.70$. Stated honestly, the acceleration is smooth, so the break location is threshold-dependent; the robust content is the acceleration together with the Chebyshev exception, not a sharp kink. [PULL] Whether the deterministic integer family’s two-regime transition, recorded separately in the prior ledger, coincides with this territory is the one comparison the paper leaves open for the next machine.

4. The instrument wall is a function, and it announces itself with one bit

The object wall is an integer: wall = support + 1, verified 300/300 exact in prior work and re-confirmed in every run here. The instrument wall is the false wall a finite precision manufactures before the object wall is reached, and the claim of this section is that it is not a random cutoff but a computable function:

$$W_{\text{instr}}(\text{dps}) = D^{-1}(\text{dps} - \kappa) + G.$$

In words: the false wall appears at the step where the object’s own cancellation profile $D(i)$ crosses the available budget, less a small headroom κ , plus a stochastic delay G contributed by the detector. The protocol was fixed before any run: calibrate κ on the random family alone; treat the integer and equispaced families as blind transfers; count a prediction correct only within ± 2 steps.

Calibration returned $\kappa \in \{1.05, -0.14, 0.14\}$ across three starved runs — that is, $\kappa = 0 \pm 1$, a headroom of essentially zero within a one-digit band. The three-sample mean prints as 0.35, and that coincidence with the framework’s fold constant $H = \pi/9$ is recorded here precisely so that no one revives it: the scatter exceeds the value, the sample is three points, and the number carries zero significant figures. [KILLED] With κ fixed on the random family, the blind transfers were:

blind transfer	dps	predicted wall	measured wall	trip type	verdict
integer 1.. M	20	18	18	sign	PASS
integer 1.. M	30	25	23	sign	PASS
equispaced $[-1,1]$	15	25	24	sign	PASS

Three for three, deltas 0, -2 , -1 , all inside the band — and the second transfer crossed the integer family’s own regime break, which is the point of using the measured profile $D(i)$ rather than a linear extrapolation. [PULL]

The mechanism finding outranks the fit. Every starved trip — three in calibration, three in transfer — fired the sign detector: the factorization stopped because a pivot went negative, not because the pivot-ratio floor was crossed. The ratio floor never fired once. **The instrument wall is announced by a single bit — the first negative pivot in the sub-floor stream.** This recovers, from the depth side and by a completely independent route, the detector law of prior work, $W_{\text{observed}} = k_{\text{onset}}(\mu, \varepsilon) + G$: the onset is deterministic, set by depth crossing budget, and the announcement waits a short stochastic interval for the first negative sign, whose spread is exactly the observed deltas. The consequence for the three-wall picture is sharp: at starved budgets the ratio floor is dead weight, and the detector wall of the prior installment is the sign channel. [PULL]

This is also, concretely, the subtraction the Riemann experiment needs. Measure $D(i)$ for an arithmetic measure at high precision, predict where the instrument wall will fall at any given budget, and count as genuine object signal only the structure that appears *in excess* of that prediction. Without this step the wall-versus- $N(T)$ comparison cannot distinguish a zero from a rounding artefact; with it, the instrument's predictable contribution can be removed before the reading is taken.

5. The affine sort: coordinate versus invariant

The read-frame is the new axis. A measure can be read from different coordinate origins, at different scales, in either orientation; the question is which quantities move with the frame and which do not. The affine group — translation, dilation, reflection — was put on the bench, and the preregistered predictions were: the object wall is invariant under all of it; the clock c is invariant under dilation and reflection (the ratio $r_{\text{out}}/\text{cap}$ cannot see them) but covariant under translation per the capacity law; and the Jacobi coefficients split, with the diagonal α carrying the translation and the off-diagonal β blind to it. Base object: integer nodes 1..30, unit weights, five frames.

frame	wall	c measured	α_0	α_1	β_1
centered ($x - 15.5$)	31	0.571	0	0	74.9167
identity (x)	31	1.191	15.5	15.5	74.9167
shift +100	31	2.461	115.5	115.5	74.9167
dilation $\times 1000$	31	1.191	15500	15500	7.4917×10^7
reflection ($-x$)	31	1.191	-15.5	-15.5	74.9167

Every entry is exact — integer nodes at four hundred digits leave no rounding residue, so the measured deviations are literally zero, not small. The object wall sat at 31 under all five frames. The clock was bit-identical under dilation and reflection, and covariant under translation exactly as $c = 2\log_{10}(r_{\text{out}}/\text{cap})$ predicts: reading the same object from a hundred units away raises the burn rate from 0.571 to 2.461 because the ratio $r_{\text{out}}/\text{cap}$ has been inflated by the shifted origin. Every α shifted by exactly the frame translation and by nothing else; every β was frame-blind under translation and reflection and scaled by exactly 10^6 under the thousand-fold dilation, the square of the length scale. [LOCKED — the covariance laws are classical; the exact confirmation is measured]

The inventory this produces is the paper’s organising result. **Object side, invariant under the affine read-group:** the wall (the rank; an integer, exact under all five frames); the off-diagonal coefficients β up to the dilation action, so β -ratios are full affine invariants; the α -gaps up to dilation, so gaps normalised by $\sqrt{\beta_1}$ are invariants. **Frame side, covariant:** the absolute α , which carry the shift and carry nothing else; the clock speed c ; the depth profile $D(i)$; the false-wall location; and every individual pivot. A confirming detail sits inside the table: $\beta_1 = 74.9167 = (M^2 - 1)/12$ is exactly the variance of the node set — the archetypal difference-quantity, translation-blind by construction, which is why it does not move when the origin does.

Two further consequences follow. The first is a geodesic frame. Writing $c(t) = 2\log_{10}(\max_i |x_i - t|/\text{cap})$ for a read origin at t , the burn rate is minimised when the origin sits at the Chebyshev centre of the support, and for any single interval the minimum is universal: $c_{\min} = 2\log_{10} 2 \approx 0.602$ digits per tick. The excess $c - c_{\min}$ is pure frame — the price of reading off-centre — and it is verified in the data, the centred integer frame measuring 0.571 against 0.602 predicted (the same arcsine offset from §3) and the far-shifted frame’s excess matching $2\log_{10}(130/14.5) = 1.905$. The second consequence is the sharpest reading of an old slogan. “Only differences survive” is now a measured sentence with a precise referent: the translation part of the Jacobi operator is a c -number, a pure additive coordinate, and all structure lives in what remains after that coordinate is subtracted. Reflection flipped the sign of every α — every label — while the wall and the β did not move: labels are frame, distinctions are object, and a bit is not a zero-or-one but a distinguishable-versus-not. What survives every admissible frame in this system is an integer, the count of distinctions — numbers as gap-labels, measured.

6. Isospectral motion: the execution layer made testable

The affine group moves the coordinates while fixing the measure. There is a second, complementary motion that fixes the *spectrum* while moving the measure — and it turns an idea that had been architecture into an experiment. The three-layer picture that emerged in dialogue distinguished representation (the coordinates: moments, α , node locations, pivots), invariant (the wall, the rank, the β -ratios — the mathematical object as an equivalence class across representations), and execution (the recurrence itself: the process that changes representation while preserving the invariants). The first two layers had been measured; the third had not, because “the process that carries the object through its coordinates” sounds unfalsifiable. The isospectral flow makes it concrete.

The Toda lattice is the canonical isospectral flow on a Jacobi operator: $dJ/dt = [J, A]$ with A the antisymmetric part of J , a Lax equation whose defining property is that the spectrum of J — here, the support of the measure — is exactly conserved for all t . By Moser’s solution the flow acts on the measure by a simple exponential reweighting, $d\mu_t \propto e^{2xt} d\mu_0$: reweight, rebuild the moments, refactor, and read. This is precisely “change the representation while preserving the invariant,” now with a dial. The preregistered predictions were: the wall is conserved (isospectral, so the count cannot move); the early clock is conserved to leading order (the support is fixed, so $r_{\text{out}}/\text{cap}$ is fixed); the representation flows (mass migrates to larger nodes, so α_0 increases, β_1 changes, $D(20)$ changes); and — the deepest rung — β_1 is *not* conserved, because Toda’s invariant is coarser than the affine sort’s object. The run, on the integer object at four representative times:

t	wall	early c	α_0	β_1	$D(20)$
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t	wall	early c	α_0	β_1	$D(20)$
0	31	1.191	15.5	74.9167	24.18
0.05	31	1.200	22.06	50.29	24.80
0.15	31	1.213	27.15	10.92	26.08
0.4	31	1.281	29.18	1.482	29.50

All four predictions held. **T1**, the wall sat at 31 at every time — the count is an exact constant of the isospectral motion. **T2**, the early clock drifted by at most 7.5% across the whole flow, inside the band, and the drift is monotone rather than noisy — which is not a blemish but the tie to §3. Toda conserves the *spectrum* exactly, so r_{out} and cap and therefore the *leading* clock constant $2\log_{10}(r_{\text{out}}/\text{cap})$ are exactly conserved; the measured 7.5% drift is the *subleading* prefactor responding to the reweighting, as mass concentrates toward $x = 30$ and the edge behaviour of the density changes. The Toda drift and the five-percent deficit of §3 are the same finite-index correction, driven by reweighting in one case and by finite index in the other. **T3**, the representation flowed exactly as constraint predicts: α_0 climbed from 15.5 to 29.18 as mass migrated to the largest node, β_1 collapsed from 74.9 to 1.48 as the variance of the concentrating measure fell, and the depth $D(20)$ moved. **T4**, β_1 was not conserved — confirming the invariant tower. [PULL for the measured drift structure; the conservation laws are classical]

Why the Toda flow is the right model of an execution layer, rather than an arbitrary deformation, is worth making explicit. The Lax form $dJ/dt = [J, A]$ is the statement that $J(t)$ is carried by a one-parameter family of orthogonal conjugations, $J(t) = Q(t)J(0)Q(t)^{-1}$, so the eigenvalues — the support — are conserved by construction while the eigenvectors, and hence the coordinates in which the operator is expressed, rotate. That is exactly the definition of an execution layer as this paper uses the term: a motion that changes the representation while preserving the invariant. The flow is completely integrable, its conserved quantities are the spectral moments $\text{tr } J^k$, and its long-time limit sorts the mass onto the extreme eigenvalue — which is why, at $t = 0.4$, α_0 has climbed almost to the largest node and β_1 has nearly collapsed. The measure is being compiled toward a point mass along a path that never disturbs the spectrum. An execution that preserves what the object *is* while moving how it is *written* is not a metaphor here; it is a named, classical, integrable flow, and the count riding invariant on top of it is the same count that rides invariant under the affine read-group.

The picture that closes the section is two groups acting on one object. The **read-group** (affine maps of the support) fixes the measure and moves the coordinates; its invariants are the wall, the β -ratios, the α -gaps. The **flow-group** (Toda) fixes the spectrum and moves the measure; its invariant is coarser — the spectrum itself, hence the wall — and it does *not* fix β_1 . The count survives both. That an integer — the number of distinctions in the object — is invariant under both a change of coordinates and a change of measure that preserves the spectrum is the strongest sense in which this program has found something coordinate-free: it is invariant under the two independent motions the structure admits.

7. Stream versus state: the path-dependence discriminator

The dialogue running alongside this work pressed a reframing: that transitions are primary and states are snapshots cut from an underlying stream — objects as compressions, the Jacobi operator as a compiled

interface to an ongoing process, the wall as the aperture limit where the instrument can no longer distinguish new structure. It is an appealing picture, and it ended with the right challenge: find an observable that distinguishes this “stream-first” reading from the conventional “state-first” one, or it stays a lens rather than a hypothesis. This section builds that observable and reports its verdict.

The naive contrast seems to trivialise in our sandbox, because the object here is a static measure and its reads commute at infinite precision. The sharper contrast is about **path-dependence below the floor**. The same mathematical pivot sequence h_i can be computed by two numerically distinct routes: Route A, an LDL^T factorization of the Hankel matrix, which forms each pivot as a Schur-complement subtraction $d_i = \mu_{2i} - \sum_{j<i} L_{ij}^2 d_j$; and Route B, the Stieltjes three-term recurrence, which builds the monic polynomials by coefficient arithmetic and computes $h_i = \langle P_i, P_i \rangle$ from moment inner products. The two routes reach the same object by entirely different arithmetic, so their rounding trajectories differ. State-first predicts they agree — the object is what it is, independent of route. Stream-first predicts that below the precision floor, where the pivot is a difference of large nearly-equal quantities, the routes diverge, because there the *execution* is doing the deciding.

The experiment ran both routes on sixteen random measures at $M = 20$, each at a per-measure starved precision chosen so that a band of upper indices falls below the budget while remaining, mathematically, strictly positive (every index below the support size has a positive pivot). Three controls were added to decompose any divergence: Route A recomputed at $dps + 1$ (a pure precision-path perturbation), and Route A recomputed with the moment summation performed in a shuffled node order (a pure input-rounding perturbation, same algorithm, same precision). The preregistered readings: S_1 , agreement of A and B at four hundred digits, testing state-first at the object tier; S_2 , disagreement below the floor, testing that path-dependence exists at all; and S_3 , the discriminator — the sub-floor sign-agreement rate between routes, high if an object component survives in the sign content, at chance if the content is pure execution.

quantity (sub-floor window, $n = 93$ signs, 16 measures)	value	95% CI
S_1 : high-precision A vs B, max relative error	8.1×10^{-378}	—
positivity of Route A below floor (truth is +)	0.527	(0.425, 0.628)
agreement A vs B (different algorithm)	0.548	(0.447, 0.650)
agreement A vs A at $dps + 1$ (precision path)	0.473	(0.372, 0.575)
agreement A vs A shuffled (summation path)	0.968	(0.932, 1.004)

The result is clean and it decomposes perfectly. **S_1** : above the floor the two routes agree to 8×10^{-378} — the object is exactly route-independent, state-first is exactly true at the object tier. **S_2** : below the floor the routes disagree far more than the 0.95 threshold — path-dependence is real at the instrument tier. **S_3 with its controls** is the substance. The object’s mathematical sign below the floor is always positive, yet Route A returns positive only 52.7% of the time, a rate whose confidence interval includes one-half: the object’s sign has been *completely washed out*. Changing the algorithm (A versus B) yields agreement 0.548, at chance. Changing the precision by a single digit (A versus A at $dps + 1$) yields agreement 0.473, at chance. But changing only the summation order of the input moments, holding algorithm and precision fixed, yields agreement 0.968 — almost perfect.

Read together, these say something precise. The sub-floor sign is **not** a property of the object: it is at chance across routes and carries none of the object's guaranteed positivity. It is **not** robust to precision: one digit flips half of it. Yet it **is** nearly deterministic given the algorithm and precision jointly, since perturbing only the input rounding barely moves it. The sub-floor sign is therefore a **deterministic fingerprint of the execution configuration** — of which algorithm ran at which precision — and it is independent of the object's mathematical value. It behaves, exactly, like a hash of the process: fixed once the process is fixed, decorrelated to chance under any change of process. [PULL]

This has a precise analogue in the program's cryptographic work, and the analogy is structural rather than decorative. The SHA-256 study treated the digest as *exhaust* — the readable output a scrambling of the process rather than the primary object — and the present result is that same structure inside the moment instrument. The sub-floor sign stream is a hash of the execution configuration in the three senses that matter: it is a deterministic function of the (algorithm, precision) pair; it is diffuse with respect to that pair, since a one-digit change in precision decorrelates it to chance exactly as a one-bit change in a hash input scrambles the output; and it is blind to the object, since the object's guaranteed positivity is invisible in it. What the channel leaks is the process that produced it, not the datum it operated on — the definition of a side channel. The refinement this forces on the program's Van Eck reading is that the leak is *timing*, not *content*: the onset of the wall is object-determined and predictable (§4), so the channel's schedule carries the object, while its payload carries only the machine. An eavesdropper on this factorization would recover the support from *when* the collapse fires and would recover only the algorithm-precision fingerprint from the bits after it. That is sharper and more defensible than "the garbage carries object geometry," and it is the version the measurement supports.

This is the answer to the threshold challenge, and it dissolves the rivalry into a tier assignment rather than declaring a winner. **State-first is exactly true of the object**: S_1 puts the two routes at 10^{-378} , the wall and the invariants are route-independent and exact. **Stream-first is exactly true of the instrument's exhaust**: the sub-floor content is pure execution, a fingerprint of the process that produced it. The stream is real — but it is the stream of the *apparatus*, not of the object, and the two live at different tiers of the same computation.

The consequence for the program's own prior claim is a genuine sharpening. The earlier work proposed, by analogy to Van Eck phreaking, that the sub-floor residue carries readable object geometry — that the exhaust leaks the object. This experiment splits that claim in two and validates the correct half. The wall's *location* — its onset — is object-determined; §4 showed it is predictable from the object's depth profile $D(i)$. The wall's *content* — the sub-floor sign stream — is execution-determined; it fingerprints the algorithm-precision pair. This is exactly what real Van Eck phreaking recovers: the emanations read the device's *process*, and the data is inferred only where process correlates with data. Here the correlation is timing, not content: the onset leaks the object (the count), the garbage after it leaks the algorithm. It also refines a prior result on the detector statistics. That the sub-floor sign stream is biased and anti-correlated rather than a fair coin was established earlier; the present decomposition shows that structure belongs to the *algorithm class*, not to the object, because across algorithms the signs decorrelate to chance while under input reshuffling within one algorithm they are preserved. The detector layer sits on the execution, not on the object — which is precisely where a three-wall separation should place it.

Two methodological points keep the reading honest. First, statistical power: at $n = 93$ paired signs the ninety-five-percent interval on an agreement rate is roughly ± 0.10 , wide enough that “at chance” means the interval contains 0.5 rather than that the point estimate is exactly 0.5 — which is the case for all three chance-level comparisons and emphatically not the case for the summation-order control at 0.968. The qualitative decomposition — two perturbations at chance, one near unity — is robust to this width; a sharper estimate of the small excess of the cross-algorithm rate over 0.5 would need more measures and is left open. Second, the summation-order control is doing specific work: it isolates the shared-algorithm rounding bias from object-instance signal. Because it holds algorithm and precision fixed and changes only the order in which the input moments are accumulated, its near-unity agreement establishes that the sign is essentially determined once the configuration is fixed, so the chance-level cross-algorithm and cross-precision rates cannot be attributed to input noise — they are genuine sensitivity of the sign to the algorithm and to the budget. Without that control, a low cross-algorithm agreement could have been dismissed as mere arithmetic jitter; with it, the conclusion that the sign is an execution-configuration fingerprint is forced.

8. The wall is a floor: reading the object from its own collapse

The preceding sections treat the wall as a boundary — §3 and §4 ask only where it falls. There is a complementary reading, and it completes the object/frame sort of §5. When the factorization hits the wall, the pivot does not merely vanish; the *direction* in which it vanishes is the object’s own defining equation. Turn the read so that the collapsing direction becomes the surface one stands on, and the wall becomes a floor. From inside the moment space there is no absolute orientation: a wall — a direction one cannot continue through — and a floor — a surface one reads from — are the same object seen with a different choice of load direction, and that choice is a frame operation. §5 sorted quantities into invariant and covariant; this section adds that even the *role* of an invariant, barrier or ground, is frame-relative, though the invariant surface itself is not. This is the program’s “side door” as arithmetic: entering a structure orthogonal to its load direction.

Two readings, one experiment, both confirmed on an $M = 8$ -point measure run to the wall, where the pivot ratio h_M/h_{M-1} collapses to 9×10^{-50} .

Reading one: the collapse reads the whole object. The polynomial P_M whose norm vanishes is monic of degree M , and its roots are the support to a maximum error of 8×10^{-51} — every node recovered exactly from the single vanishing pivot. Equivalently, the coefficient vector of P_M is the null vector of the singular Hankel matrix, with $H_M \cdot \text{coeff}(P_M) = 0$ to 2×10^{-60} relative to scale. The collapse direction is neither noise nor an endpoint; it is the object’s defining constraint $P_M(x) = \prod_i (x - x_i) = 0$, the characteristic polynomial of the Jacobi matrix. Standing on the wall, one reads the entire support at once — the wall is the point of maximum information about the object, not the point where information ends. [LOCKED — that the orthogonal-polynomial roots are the quadrature nodes and Jacobi eigenvalues is classical; the reframing of the collapse as a readout surface is the new content]

Reading two: turn the field. The power-moment reading $\mu_k = \int x^k d\mu$ — the vertical direction — walls out at M because the discrete measure has exactly M distinctions to give. Rotate ninety degrees to the resolvent reading $S(z) = \int d\mu/(z - x) = \sum_i w_i/(z - x_i)$, and the same object does not wall: it is an analytic function with exactly M poles. The Jacobi continued fraction built from the measured α, β terminates at level M and

reproduces the resolvent to a maximum relative error of 9×10^{-58} at three test points, and its denominator is exactly P_M , so its poles are the nodes. The wall of the polynomial reading and the pole structure of the resolvent reading are the same polynomial seen with the field turned: where the norm collapses to zero (the wall) is where the resolvent's poles stand (the floor one reads the spectrum from). [confirmed; the resolvent-continued-fraction identity is classical]

This is also the deepest statement of the program's constraint-first ontology, and it belongs directly to the apex of §9. The object *is* its wall polynomial — the support is the zero-set of P_M , and P_M is precisely what the collapse hands over. "Constraint is prior to the object" is here literal: the defining constraint $P_M = 0$ is not derived from the object, it is the collapse, and the object — the nodes — is its solution set. The wall is where the object states its own equation. The reading sharpens the finite-to-infinite passage of §10 as well: at finite support the wall polynomial is a finite product providing a floor of M poles, and as support grows the wall of level M is the floor of the next — the vanishing polynomial at each stage is the ground from which the next is read — so in the limit the discrete poles become the continuous spectrum, the count becomes a density, and the floor becomes the counting function itself.

9. The constraint apex: sorting the ontology

Across the dialogue a single reframing kept surfacing, and the measurements have now earned it a place as the organising principle rather than a flourish. The recurrence never *chooses*. At each step it eliminates what cannot continue. The state-then-transition picture — a state executes to produce the next state — is replaced by a constraint picture: a space of possible continuations, a constraint, and the one survivor. Execution is constraint becoming visible.

Every result in this paper is that picture in a different guise. The pivot is slack: the truncated moment problem confines the next even moment to a half-line whose boundary is exactly the subtracted mass, so the pivot $h_i = \mu_{2i} - \mu^*$ is literally the width of the remaining admissible freedom. The cancellation depth $D(i)$ is the freedom already consumed; the clock c is the rate at which constraint resolves; the wall is the step at which the admissible set collapses to a point — enforced, not discovered, and enforced by algebra, since both sides of that identity are computed by the same decomposition and no numerical check could add anything. The affine sort is constraint-first too: one does not define the invariant, one asks what survives every admissible transformation and lets everything else disappear. And the compiled-state result is the same statement — the Jacobi operator is not storing history by design; it is the minimal object still consistent with every prior constraint, everything else having been eliminated. Compression is constraint.

This recasts the program's oldest slogan. "Today is the sum of all yesterdays" becomes, more precisely, **today is everything yesterday could not eliminate** — and in this system that is exactly the truncated moment problem, the present state being the intersection of all past constraints, collapsing to a point at the wall. It also gives "realization, not search" a concrete referent: the recurrence does not try possibilities and select among them; it reads the only branch still consistent with the accumulated constraints. There is no free search because the constraints have already determined the continuation; execution simply follows it until there is no continuation left.

The dialogue's own refinement of the invariant deserves recording, because it is right and it has a classical name. The object is not literally "what survives"; it is the equivalence class the constraints define, and what survives in any given reading is merely the representative that frame exposes. The representative-free coordinates already exist in the literature as **canonical moments** (Dette–Studden): each records where a moment sits inside its admissible range given all prior constraints, a coordinate built from slack and invariant under affine maps of the support. That is the exact formal content of "the object is the equivalence class, not the survivor," and it is the natural coordinate system for the invariant layer §5 measured.

It is worth stating what the constraint apex does and does not claim about search, because "realization, not search" is easy to over-read. The claim is not that no computation occurs — the factorization plainly computes, tick by tick, burning the digits §3 counts. The claim is narrower, and it is exactly what the wall-as-floor result of §8 exhibits: at no step does the recurrence choose among admissible continuations, because at each step exactly one continuation is consistent with the accumulated constraints, and the wall is where that one continuation is the zero polynomial. The pivot measures how much freedom remains, the recurrence spends it, and when it reaches zero the object has stated its own defining equation. This is realization in the strict sense — the unique permitted continuation is read off, not selected — and it is why the same object reached by two different algorithms agrees to 10^{-378} above the floor (§7): there was never a branch point at which the two algorithms could have realized different objects. Search would require branch points; the moment recurrence has none until the wall, and at the wall it has run out of object rather than run out of options. The execution layer is real, but it is deterministic all the way down, and its only genuine freedom — the sub-floor sign — belongs to the machine, not the mathematics. That is the whole of the constraint-first ontology stated operationally: the object is the fixed point of its constraints, execution is the reading of the one branch those constraints permit, and the only place chance enters is below the floor, in the exhaust, where the object has already been fully determined and only the apparatus is still speaking.

The architecture that results is not a stack but a fork. Constraint is the apex. From it descend, as two consequences of one structure, the *invariants* (what the constraint preserves under transformation — the wall, the rank, the β -ratios) and the *execution* (the process that realises the unique permitted continuation — the isospectral flow, the recurrence, the factorization sweep). Below both sits *representation* — the coordinates any particular frame exposes, which change under both the read-group and the flow-group. The measured shadow of the execution layer is real but modest: the invariants are route-independent (§7, S_1 at 10^{-378}) while the cost is frame-dependent (§5, the same object charging 0.571 digits per tick from its centre and 2.461 from a hundred units away). Whether execution deserves an ontological standing distinct from invariant and representation is beyond this instrument; it is recorded as architecture, PULL, not as measurement.

Three items from the same dialogue are recorded here so they do not drift into the record uncorrected. "Nothing is stored" is the mirror image of an already-refuted claim and is itself refuted: storage is not absent but global — nothing is stored locally, and every moment $\mu_k = \sum_i w_i x_i^k$ touches every node, so a distinction anywhere shifts every coefficient. [the local-storage picture is REFUTED; global coupling is the established statement] "Every object is part of every other object's frame," the mutual-measurement graph, has no shadow in this sandbox: the read here is one-way and back-action-free, the object shaping the instrument while the instrument never touches the object, so the symmetric graph is a physics hypothesis, not an in-system result. [Ω] And the reading of light as a different relationship to the stream has no contact point with anything measured. [Ω] Keeping these on the shelf, labelled, is the same discipline that let the earlier

“frozen crystal interior” metaphor be caught by the code before it calcified: an elegant compression that a measurement refuses is a result, and the refusal is the thing worth keeping.

The honest boundary, restated where it belongs. None of this licenses a claim about the physical universe. What it does is convert “stream versus state” from a matter of taste into a measurable property — path-dependence of the exhaust below the floor — that any physical system could in principle be interrogated for. The sandbox has not shown that reality is a constraint stream; it has shown that the question has a definite answer *for this object*, that the answer is a clean tier assignment, and that the method for extracting it is a preregistered measurement rather than an argument. That is the threshold at which a lens becomes a hypothesis, and it is as far as the instrument reaches.

10. The invariant at infinite support, and the Clay path

At finite support the terminal invariant is a count: $\text{wall} = \text{support} + 1$, exact 300/300 in prior work and affine-proof here, and — from §6 — conserved under isospectral flow as well. At infinite support the count disappears, and the question is what replaces it. The dialogue’s proposal, which the finite results make plausible, is the *topology of distinguishability*: for a finite-support measure this reduces to a count, but for an infinite-support measure the invariant must become something like a spectral type, an essential support, or a counting function — a count that has learned to be a function of a continuous parameter.

The Clay-adjacent experiment is exactly that passage. For prime-derived moments, the question “does the wall track $N(T)$?” is the question of whether the count invariant becomes the zero-counting function — whether the integer that survives every frame at finite support becomes, at infinite support, a function of height along the critical line. The contribution of this paper to that experiment is the subtraction that makes it readable. §3 gives the object’s cancellation profile $D(i)$ and its capacity clock; §4 turns that profile plus the budget into a prediction of the instrument wall; and the reading of $N(T)$ can then be taken on the residual, the structure in excess of the predicted instrument wall, rather than on a raw wall that conflates object and apparatus. The run remains gated on two items before its numbers are trustworthy: the precision of the build-order run that produced the earlier 10^{-48} figure, and agreement on the moment convention between instances.

The next bridge is nearer and testable now. The clock speed c has been measured only for geometric families, where it is a conformal invariant of the support and nothing more. Measured instead on the a_p moments of the four Birch–Swinnerton-Dyer curves already reproduced in the Scale Notebook, c becomes an *arithmetic* observable, and the question is whether it carries arithmetic content — whether the burn rate of a curve’s moment factorization knows its conductor or its rank. That experiment needs no new theory, only the same profile run on an arithmetic measure, and it is the natural continuation of the anchor this paper opened with.

11. Status ledger

claim	status	evidence
Hankel positivity $\Leftrightarrow$ existence of	LOCKED (classical)	Haviland; necessity from sum-of-squares

claim	status	evidence
measure		
determinacy $\Leftrightarrow$ least Hankel eigenvalue $\rightarrow 0$	LOCKED (classical)	conditioning = moment geometry
wall = support + 1	LOCKED (prior)	300/300 exact; re-confirmed every run
capacity clock $c = 2\log_{10}(r_{\text{out}}/\text{cap})$	PULL	4/4 within 10%
–5% deficit is arcsine Szegő prefactor	PULL	equilibrium fit $c = 0.602$ (0.1%), $\gamma = -0.47 \approx -1/2$
Gauss–Chebyshev exactness $\Rightarrow$ one-clock measure	LOCKED (classical)	residual 0.106 digits, $i = 3..29$
gear change = constrained-equilibrium surcharge	PULL	integer slope 1.19 $\rightarrow$ 2.24; break $t \approx 0.70$, threshold-dependent
instrument wall $W = D^{-1}(\text{dps} - \kappa) + G, \kappa = 0 \pm 1$	PULL	3/3 blind transfer, ± 2 band
sign-first announcement (detector = sign channel)	PULL	6/6 starved trips; ratio floor never fired
affine covariance (α shifts, β blind, wall fixed)	LOCKED (classical)	exact zeros, five frames
Toda: wall and leading clock conserved; β_1 not	PULL / LOCKED	wall = 31 all t ; 7.5% subleading drift; $\beta_1: 74.9 \rightarrow 1.5$
object is state-first and route-exact	LOCKED (measured)	S1: A vs B to 8×10^{-378}
wall is a floor: collapse = object's defining polynomial	LOCKED (classical); reframe PULL	P_M roots = nodes to 8×10^{-51} ; J-fraction = resolvent to 9×10^{-58}
sub-floor sign = execution fingerprint, not object	PULL	positivity 0.527; cross-route 0.548; cross-precision 0.473; cross-summation 0.968
Van Eck refined: onset = object, sign-content = process	PULL	§4 onset predictability + §7 decomposition
strong relationalism (no coordinate-free quantities)	REFUTED	wall exact under five frames and isospectral flow
$\bar{\kappa} = 0.35$; 0.35 in enumeration clock rates	KILLED / ABSENT	3-sample mean, scatter ± 1 ; no fitted slope in (0.33,0.37)
constraint apex; execution as distinct layer	PULL (architecture)	route-independence of invariants vs frame-dependence of cost
local storage absent	REFUTED	global coupling: μ_k touches every node
mutual-measurement graph; light as stream relation	Ω	no in-system shadow (one-way, back-action-free read)

12. Reproduction and provenance

Every number in this paper was regenerated from moments within a single session; nothing was trusted from earlier output. Seven scripts accompany it: the four-family clock and starvation runs; the translation-covariance frame test; the five-frame affine sort with Jacobi data; the Toda isospectral flow; the path-dependence discriminator with its three controls; the subleading-correction refit; and the wall-is-a-floor reconstruction (null polynomial and terminating resolvent). All run under mpmath at four hundred digits for object-side profiles, with each PASS/FAIL preregistered in the text that precedes it. Per the program's standing discipline, no PULL row is promoted to LOCKED on the strength of this document; promotion waits on an independent re-run, preferably on separate hardware and by a separate model instance. Two specific items are outstanding for the next machine: whether the deterministic integer family's two-regime transition coincides with the gear-change territory near $t \approx 0.70$, and the precision of the build-order run underlying the earlier moment-accuracy figure.

Provenance. Produced within the multi-AI cross-validation network. The three-layer articulation, the constraint apex, the canonical-moments identification, and the topology-of-distinguishability generalization entered through the GPT node across several rounds; the isospectral-flow and path-dependence experiment designs, the runs, and the sorting verdicts are the Claude node's; direction, standards, and the anchor discipline are the author's. The rigorous asymptotics reference supplied the classical scaffolding of §2 and the Szegő prefactor that resolved §3.

Inside the Geometry: The Pythagorean Constraint Behind the Shape Channel

Christoffel Rank Read program — Clay bridge notebook II · Dean Kulik, QuHarmonics

We are inside the geometry, so the ordinary rules are in force — including the Pythagorean theorem, which here is not a metaphor. For an elliptic curve the Frobenius eigenvalues at a good prime p are $\sqrt{p}e^{i\theta_p}$ and $\sqrt{p}e^{-i\theta_p}$, and the trace is $a_p = 2\sqrt{p}\cos\theta_p$. Two facts follow, and they are the whole notebook:

$$\cos^2 \theta_p + \sin^2 \theta_p = 1 \Leftrightarrow a_p^2 + (\sqrt{4p - a_p^2})^2 = (2\sqrt{p})^2.$$

The left is the unit circle; the right is a literal Pythagorean triple with legs a_p (the real trace) and $\sqrt{4p - a_p^2}$ (the imaginary co-trace) and hypotenuse $2\sqrt{p}$. This is exactly the **Weil bound** $|a_p| \leq 2\sqrt{p}$ — the **Riemann Hypothesis for the curve over F_p** , which is *proven* (Hasse). The support $[-1, 1]$ of $x_p = a_p/(2\sqrt{p})$ that fixed the shape-channel clock to $2\log_{10} 2$ in the previous notebook is the *shadow of this Pythagorean circle*. The shape channel was reading the local Riemann Hypothesis all along.

The measure of $\{x_p\}$ splits into two orthogonal legs, and the split is where the Clay problems separate:

- the **symmetric leg** (even moments) — the semicircle shape, universal across all non-CM curves, blind to rank;
- the **antisymmetric leg** (odd moments) — the reflection asymmetry, which carries a faint rank bias that arithmetic weighting amplifies.

"Context + reflection = truth," in the program's older phrasing: the symmetric context is universal, the reflection carries the number.

Preregistered.

- P1 (local RH / unit circle): every $x_p \in [-1, 1]$; zero violations, all four curves.
- P2 (symmetric leg universal): even moments match the semicircle values $C_k/4^k$ and agree across ranks to within a few percent.

- P₃ (antisymmetric leg carries rank): the odd-moment asymmetry grows with rank (weak, unweighted; sharpened by the $\log p/p$ weighting of the prior notebook).

Discipline: Weil/Hasse is a theorem (LOCKED); the semicircle moments are the Sato–Tate law (LOCKED for these non-CM curves); the rank content of the asymmetry is suggestive (PULL). The cross-Clay map at the end is a lens, explicitly not a claim of solution.

```
# --- setup: point-counted a_p for four curves, one per rank (all non-CM) ---
import mpmath as mp
import math, time

def primes_upto(n):
    s = bytearray([1]) * (n + 1); s[0] = s[1] = 0
    for i in range(2, int(n**0.5) + 1):
        if s[i]: s[i*i::i] = bytearray(len(s[i*i::i]))
    return [i for i in range(2, n + 1) if s[i]]

def ap_fast(cv, p):
    # a_p = p + 1 - #E(F_p); affine points counted via a QR table, O(p) per prime
    a1, a2, a3, a4, a6 = cv
    if p == 2:
        cnt = sum(1 for x in range(2) for y in range(2)
            if (y*y + a1*x*y + a3*y - (x**3 + a2*x*x + a4*x + a6)) % 2 == 0)
        return 2 + 1 - (cnt + 1)
    is_sq = bytearray(p)
    for t in range(p): is_sq[(t*t) % p] = 1
    cnt = 0
    for x in range(p):
        rhs = (x*x*x + a2*x*x + a4*x + a6) % p
        b = (a1*x + a3) % p
        disc = (b*b + 4*rhs) % p
        if disc == 0: cnt += 1
        elif is_sq[disc]: cnt += 2
    return p - cnt

curves = {'11a1': ((0, -1, 1, -10, -20), 11, 0),
          '37a1': ((0, 0, 1, -1, 0), 37, 1),
          '389a1': ((0, 1, 1, -2, 0), 389, 2),
          '5077a1': ((0, 0, 1, -7, 6), 5077, 3)}
```

```

P = 20000
pr = primes_upto(P)
to = time.time()
apdata = {n: ([p, ap_fast(cv, p)] for p in pr if N % p != 0), N, r)
    for n, (cv, N, r) in curves.items()}
print(f"a_p over {len(pr)} primes up to {P} in {time.time()-to:.1f}s")

```

a_p over 2262 primes up to 20000 in 22.7s

Part 1 — the Pythagorean support: Weil = local RH = unit circle

Check the literal Pythagorean statement: every normalized trace $x_p = a_p/(2\sqrt{p})$ lies in $[-1,1]$, i.e. the Frobenius eigenvalues lie on the circle of radius $\sqrt{p}$. Zero violations is the Riemann Hypothesis over F_p , holding at every good prime.

```

# --- Part 1: verify the unit-circle / Weil bound; exhibit the per-prime triple ---
print("local RH (Weil): x_p = a_p/(2 sqrt p) in [-1,1] ?\n")
print(f" {curve:8s} {rank:6s} {max|x_p|:12s} {violations:12s} {verdict}")
for n, (xs, N, r) in apdata.items():
    xmax = max(abs(ap))/(2*math.sqrt(p)) for (p, ap) in xs
    viol = sum(1 for (p, ap) in xs if abs(ap) > 2*math.sqrt(p) + 1e-9)
    print(f" {n:8s} {r:<6d} {xmax:<12.6f} {viol:<12d} {'HOLDS' if viol==0 else 'FAIL'}")

# the Pythagorean triple per prime: a_p^2 + (sqrt(4p - a_p^2))^2 = (2 sqrt p)^2
print("\nliteral Pythagorean triple (11a1, first good primes): a_p^2 + co^2 = 4p")
xs = apdata['11a1'][0]
for (p, ap) in xs[:6]:
    co2 = 4*p - ap*ap
    print(f" p={p:3d}: a_p={ap:3d} co=sqrt(4p-a_p^2)={math.sqrt(co2):7.3f} "
          f"a_p^2+co^2={ap*ap+co2:5d} 4p={4*p:5d}")
print("\nP1: the unit circle is the exact constraint fixing the shape support to [-1,1].")

```

local RH (Weil): $x_p = a_p/(2 \sqrt{p})$ in $[-1,1]$?

```

curve rank max|x_p| violations verdict
11a1 0 0.994020 0 HOLDS
37a1 1 0.996484 0 HOLDS
389a1 2 0.993301 0 HOLDS
5077a1 3 0.996056 0 HOLDS

```

literal Pythagorean triple (11a1, first good primes): $a_p^2 + c^2 = 4p$

p= 2: a_p= -2 co=sqrt(4p-a_p^2)= 2.000 a_p^2+c^2= 8 4p= 8

p= 3: a_p= -1 co=sqrt(4p-a_p^2)= 3.317 a_p^2+c^2= 12 4p= 12

p= 5: a_p= 1 co=sqrt(4p-a_p^2)= 4.359 a_p^2+c^2= 20 4p= 20

p= 7: a_p= -2 co=sqrt(4p-a_p^2)= 4.899 a_p^2+c^2= 28 4p= 28

p= 13: a_p= 4 co=sqrt(4p-a_p^2)= 6.000 a_p^2+c^2= 52 4p= 52

p= 17: a_p= -2 co=sqrt(4p-a_p^2)= 8.000 a_p^2+c^2= 68 4p= 68

P1: the unit circle is the exact constraint fixing the shape support to [-1,1].

Part 2 — the two legs: symmetric shape (universal) and antisymmetric reflection (carries rank)

Decompose the measure of $\{x_p\}$ by parity. The **even** moments $m_{2k} = \langle x^{2k} \rangle$ are the symmetric leg; for the Sato–Tate semicircle they equal the Catalan values $C_k/4^k$, and they are universal — the same for every non-CM curve, blind to rank. The **odd** moments m_{2k+1} are the antisymmetric (reflection) leg; the semicircle is symmetric, so they vanish in the limit, and their small residual carries a faint bias that grows with rank — the unweighted shadow of the arithmetic signal the *log p/p* weighting amplified into the Nagao ladder.

```

# --- Part 2: symmetric leg (even moments vs Catalan) and antisymmetric leg (odd moments) ---
from math import comb
def catalan(k): return comb(2*k, k)/(k+1)

theory = [catalan(k)/4**k for k in range(1, 5)]
print("symmetric leg -- even moments m_2k vs semicircle C_k/4^k")
print(" theory: ", [f"{t:.4f}" for t in theory], "\n")
print(f" {curve':8s}{rank':6s}{even moments (m2,m4,m6,m8)':34s}")
even_by_rank = {}
for n, (xs, N, r) in apdata.items():
    x = [ap/(2*math.sqrt(p)) for (p, ap) in xs]; M = len(x)
    even = [sum(v**(2*k) for v in x)/M for k in range(1, 5)]
    even_by_rank[r] = even[0]
print(f" {n:8s}{r:<6d}{str([f'{e:.4f}' for e in even])}")

```

```

spread_m2 = max(even_by_rank.values()) - min(even_by_rank.values())
print(f"\n m2 spread across ranks: {spread_m2:.4f} -> "
      f"{'PASS (universal, rank-blind)' if spread_m2 < 0.01 else 'CHECK'}")

print("\nantisymmetric leg -- odd moments m_2k+1 (reflection asymmetry)")
print(f" {'curve':8s}{'rank':6s}{'odd moments (m1,m3,m5,m7)':34s}")
m1_by_rank = {}
for n, (xs, N, r) in apdata.items():
    x = [ap/(2*math.sqrt(p)) for (p, ap) in xs]; M = len(x)
    odd = [sum(v**(2*k+1) for v in x)/M for k in range(0, 4)]
    m1_by_rank[r] = odd[0]
    print(f" {n:8s}{r:<6d}{str([f'{o:+.4f}' for o in odd])}")
vals = [m1_by_rank[r] for r in (0, 1, 2, 3)]
print(f"\n m1 by rank 0..3: {[round(v,4) for v in vals]}")
print(f" trend (r=2,3 clearly below r=0,1): "
      f"{'PRESENT — asymmetry grows with rank' if vals[3] < vals[2] < vals[0] else 'weak'}")
print(" (unweighted asymmetry is faint; the log p / p weighting sharpens it into the rank ladder)")

```

```

symmetric leg -- even moments m_2k vs semicircle C_k/4^k
theory:  ['0.2500', '0.1250', '0.0781', '0.0547']

curve rank even moments (m2,m4,m6,m8)
11a1  0  ['0.2463', '0.1216', '0.0750', '0.0517']
37a1  1  ['0.2447', '0.1199', '0.0741', '0.0516']
389a1 2  ['0.2511', '0.1262', '0.0793', '0.0558']
5077a1 3  ['0.2478', '0.1222', '0.0753', '0.0520']

m2 spread across ranks: 0.0064 -> PASS (universal, rank-blind)

antisymmetric leg -- odd moments m_2k+1 (reflection asymmetry)
curve rank odd moments (m1,m3,m5,m7)
11a1  0  ['+0.0030', '+0.0037', '+0.0023', '+0.0015']
37a1  1  ['-0.0015', '-0.0024', '-0.0014', '-0.0009']
389a1 2  ['-0.0157', '-0.0094', '-0.0065', '-0.0051']
5077a1 3  ['-0.0286', '-0.0174', '-0.0111', '-0.0078']

m1 by rank 0..3: [0.003, -0.0015, -0.0157, -0.0286]
trend (r=2,3 clearly below r=0,1): PRESENT — asymmetry grows with rank
(unweighted asymmetry is faint; the log p / p weighting sharpens it into the rank ladder)

```

Reading Part 2 — the Pythagorean split of the invariants

The symmetric leg is the **shape**: semicircle, Catalan moments, identical across ranks. It is the universal, state-first layer — the object as a symmetry class, fixed by the unit-circle constraint. The antisymmetric leg is the **reflection**: it should vanish by symmetry, and it nearly does, but its residual tilts with rank. Rank is not in the shape; it is in the *asymmetry* of the reflection, and the arithmetic weighting is the lens that turns that whisper into a clean readout.

This is "shape before value" as a Pythagorean decomposition on real arithmetic: $\text{shape}^2 \perp \text{reflection}^2$, the symmetric leg carrying the universal geometry and the antisymmetric leg carrying the number. The critical-line balance of the shape channel — the reason the odd moments *want* to be zero — is the conservation-law flavor the Riemann Hypothesis carries: at the point of perfect symmetry there is no gradient, and every deviation is a signal, not a defect.

The cross-Clay lens — a method, with its anchors and its open parts marked

The recurring operation across every extension in this program is one move: **when a reading saturates, rotate to an orthogonal reading that preserves the object and exposes a different invariant**. Stated as a procedure —

1. identify the invariant the current representation exposes;
2. detect where it loses resolving power (the wall);
3. construct an orthogonal reading of the same object exposing a different invariant;
4. check the two readings agree on their overlap.

Through this lens the Clay problems line up along two axes, and it is worth being exact about what is proven and what is open. This is an organizing map, **not** a claim of solution.

Existence vs reachability (the SHA principle: constraints forbid the *path*, not the *state*).

- *P* vs *NP* is the pure form: a solution to an NP problem provably *exists* in the search space; the question is whether a feasible (polynomial) read-path reaches it. Open. This is exactly the SHA-collision statement — the state is there, the trajectory may not be.

Shape vs value, balanced by a Pythagorean constraint (this notebook's structure).

- *Riemann Hypothesis*. The shape channel is the critical line — the symmetric-balance / unit-circle constraint one level up from Weil. The *local* RH (Weil/Hasse, the unit circle verified in Part 1) is **proven**; the *global* RH is the same balance for the completed ζ — open. The value channel is the explicit-formula weighting that reads the zeros.
- *Birch–Swinnerton-Dyer*. The shape channel (Part 1–2) is the Sato–Tate symmetry class, rank-blind and universal, with local RH proven. The value channel (prior notebook) is the analytic weighting that reads the rank. The rank formula is open; the reading structure is verified.

Walls and gaps in a spectral measure.

- *Yang–Mills mass gap*. A mass gap is a gap between 0 and the bottom of the spectrum of the Hamiltonian — a support-edge, a wall in a spectral measure, in the same family as the Christoffel wall. Open; the analogy is structural, labeled as such.
- *Navier–Stokes regularity*. Blow-up versus smoothness is whether the representation saturates in finite time — a wall in the time axis rather than the moment index. The connection is a loose analogy, flagged as speculative.
- *Hodge conjecture*. Which cohomology classes are algebraic is a question of which invariants are legible in which reading — a shape/value visibility problem. Analogy, labeled.

What is actually established here. The unit-circle / Weil structure (Part 1) is a proven anchor and it is the local Riemann Hypothesis. The semicircle shape (Part 2) is the Sato–Tate law. The rank-in-the-asymmetry observation is suggestive and belongs with the Nagao heuristic (PULL). The value of the lens is not any single result; it is that **the same read-rotation procedure keeps exposing structure where a previous coordinate system had genuinely saturated** — across orthogonal polynomials, structured matrices, elliptic curves, and spectral problems. The strongest evidence for the method will be its continued success across unrelated domains, not a claim about any one of them. The next concrete axis on the Clay side is the same two-reading split on the prime-Stieltjes measure for ζ — a shape reading (the wall as conformal invariant / self-measuring counter) and an arithmetic-weighted reading (the wall tracking $N(T)$) — which remains gated on the moment-convention alignment between instances.

The Side Door on BSD: Two Readings of a_p , and the Axis That Reveals Rank

Christoffel Rank Read program — BSD bridge notebook · Dean Kulik, QuHarmonics

Mindset. When a reading hits a wall — when the current plane runs out of degrees of freedom — the move is not to stop but to *add an axis*, the way a machine out of RAM introduces virtual memory instead of halting.

This notebook runs that move on real arithmetic, inside one of the Clay gravity wells (Birch–Swinnerton-Dyer). The same local data a_p — the point counts of an elliptic curve reduced mod p — is read two ways:

1. as a **symmetry measure**: the normalized $x_p = a_p/(2\sqrt{p}) \in [-1,1]$ as the support of a distribution, whose moment-factorization *clock* we measure;
2. as an **arithmetic sum**: the same a_p weighted by $\log p/p$ — the cheap, real-computable shadow of the behavior of $L(E, s)$ at $s = 1$.

The first reading should be **blind to rank**. For a non-CM curve the x_p obey the Sato–Tate (semicircle) law regardless of rank, so the clock reads the symmetry *type* and nothing finer. The second reading should **see rank**: by the explicit formula the p -weighted sum tracks the order of vanishing of $L(E, s)$ at $s = 1$, which BSD identifies with the Mordell–Weil rank. Rank is not in the raw distribution; it lives on the added axis — the analytic continuation.

Four curves, one per rank, all non-CM: **11a1** ($r = 0$), **37a1** ($r = 1$), **389a1** ($r = 2$), **5077a1** ($r = 3$).

Preregistered predictions.

- **P1 — rank-blind geometry**. The measured clock c agrees across all four ranks. PASS if the spread is < 0.05 .
- **P2 — rank-visible arithmetic**. The Nagao sum is strictly monotone increasing in rank at the largest budget and roughly tracks it.

Standing discipline: every number is regenerated from a_p ; Sato–Tate universality is a theorem for these (non-CM) curves — LOCKED; the Mestre–Nagao rank statistic is a heuristic — PULL; nothing is self-validated.

```
# --- setup: primes, point-counting a_p, the four curves (one per rank) ---
import mpmath as mp
import math, time

def primes_upto(n):
    s = bytearray([1]) * (n + 1); s[0] = s[1] = 0
```

```

for i in range(2, int(n**0.5) + 1):
    if s[i]: s[i*i::i] = bytearray(len(s[i*i::i]))
return [i for i in range(2, n + 1) if s[i]]

def ap_fast(cv, p):
    # a_p = p + 1 - #E(F_p) for  $y^2 + a_1 xy + a_3 y = x^3 + a_2 x^2 + a_4 x + a_6$ 
    # counts affine points via a quadratic-residue table, O(p) per prime
    a1, a2, a3, a4, a6 = cv
    if p == 2:
        cnt = sum(1 for x in range(2) for y in range(2)
                    if (y*y + a1*x*y + a3*y - (x**3 + a2*x*x + a4*x + a6)) % 2 == 0)
        return 2 + 1 - (cnt + 1)
    is_sq = bytearray(p)
    for t in range(p): is_sq[(t*t) % p] = 1
    cnt = 0
    for x in range(p):
        rhs = (x*x*x + a2*x*x + a4*x + a6) % p
        b = (a1*x + a3) % p
        disc = (b*b + 4*rhs) % p      # complete the square:  $(2y+b)^2 = b^2 + 4 rhs$ 
        if disc == 0: cnt += 1
        elif is_sq[disc]: cnt += 2
    return p - cnt      # = p + 1 - (#affine + 1)

curves = {'11a1': ((0, -1, 1, -10, -20), 11, 0),
          '37a1': ((0, 0, 1, -1, 0), 37, 1),
          '389a1': ((0, 1, 1, -2, 0), 389, 2),
          '5077a1': ((0, 0, 1, -7, 6), 5077, 3)}

P = 20000
pr = primes_upto(P)
to = time.time()
apdata = {name: ([p, ap_fast(cv, p)] for p in pr if N % p != 0), N, r)
            for name, (cv, N, r) in curves.items()}
print(f"a_p computed for {len(pr)} primes up to {P} in {time.time()-to:.1f}s")
for name, (xs, N, r) in apdata.items():
    print(f" {name:7s} conductor {N:5d} rank {r} good primes used: {len(xs)}")

```

```

a_p computed for 2262 primes up to 20000 in 25.0s
11a1 conductor 11 rank 0 good primes used: 2261
37a1 conductor 37 rank 1 good primes used: 2261
389a1 conductor 389 rank 2 good primes used: 2261

```

Part 1 — a_p as a symmetry measure: the clock is rank-blind

Read $x_p = a_p/(2\sqrt{p}) \in [-1,1]$ as the support of a measure with unit weights, and form its Hankel moment matrix $H_{ij} = \mu_{i+j}$, $\mu_k = \sum_p x_p^k$. The Cholesky sweep of H has pivots h_i (the monic orthogonal-polynomial norms), and the **cancellation depth** $D(i) = \log_{10}(\mu_{2i}/|h_i|)$ counts the digits destroyed to resolve pivot i . Its early slope is the **clock** $c = dD/di$, which the capacity law predicts to be $c = 2\log_{10}(r_{\text{out}}/\text{cap})$ with $\text{cap} = 1/2$ for the interval $[-1,1]$.

For the Sato–Tate law this is universal across non-CM curves — the support is $[-1,1]$ and the density is the semicircle regardless of rank. If rank were visible in the distribution, the four clocks would separate.

```
# --- Part 1: cancellation-depth clock of the Sato-Tate measure, per curve ---
mp.mp.dps = 120
nmax = 26

def depth_profile(pts, nmax):
    mus = [mp.fsum((x**k if k > 0 else mp.mpf(1)) for x in pts) for k in range(2*nmax)]
    L = [[mp.mpf(0)]*nmax for _ in range(nmax)]; d = [mp.mpf(0)]*nmax
    for i in range(nmax):
        s = mus[2*i] - mp.fsum(L[i][j]*L[i][j]*d[j] for j in range(i)); d[i] = s
        L[i][i] = mp.mpf(1)
        if s != 0:
            for k in range(i+1, nmax):
                L[k][i] = (mus[i+k] - mp.fsum(L[k][j]*L[i][j]*d[j] for j in range(i)))/s
    return [float(mp.log(abs(mus[2*i])/abs(d[i]))/mp.log(10)) if d[i] != 0 else float('inf'))
            for i in range(nmax)]

def slope(dep, lo, hi):
    xs = list(range(lo, hi+1)); ys = [dep[i] for i in xs]; n = len(xs)
    mx = sum(xs)/n; my = sum(ys)/n
    return sum((x-mx)*(y-my) for x, y in zip(xs, ys))/sum((x-mx)**2 for x in xs)

print("clock of the a_p/(2 sqrt p) measure (cap = 1/2 for [-1,1]):\n")
print(f" {curve':8s}{rank':6s}{r_out':9s}{lead c':9s}{measured c':12s}")
clocks = {}
for name, (xs, N, r) in apdata.items():
    pts = [mp.mpf(ap)/(2*mp.sqrt(p)) for (p, ap) in xs]
```

```

dep = depth_profile(pts, nmax)
c = slope(dep, 3, 18)
r_out = max(abs(x) for x in pts)
lead = float(2*mp.log(r_out/(mp.mpf(1)/2))/mp.log(10))
clocks[name] = c
print(f" {name:8s}{r:<6d}{float(r_out):<9.4f}{lead:<9.3f}{c:<12.3f}")

spread = max(clocks.values()) - min(clocks.values())
print(f"\nsread across ranks 0..3: {spread:.4f}")
print(f"P1 (rank-blind, spread < 0.05): {'PASS' if spread < 0.05 else 'FAIL'}")

```

clock of the $a_p/(2 \sqrt{p})$ measure (cap = 1/2 for [-1,1]):

```

curve rank r_out lead c measured c
11a1 0 0.9940 0.597 0.539
37a1 1 0.9965 0.599 0.537
389a1 2 0.9933 0.596 0.541
5077a1 3 0.9961 0.599 0.538

spread across ranks 0..3: 0.0037
P1 (rank-blind, spread < 0.05): PASS

```

Reading Part 1 — this is the wall

The four clocks are identical to a few thousandths, across ranks 0, 1, 2, 3. Both the leading rate and the sub-leading deficit (the clocks sit below the leading ~ 0.60 by the same amount — the Sato–Tate semicircle's soft-edge prefactor) are the *same* for every curve. The moment geometry of a_p reads the **Sato–Tate symmetry class** and is blind to rank, and to leading order to the conductor.

This is a *wall* in the program's sense: the real-line measure reading has no remaining degree of freedom in which to express rank. It is also the wall-is-a-floor result on arithmetic data — standing on this measure's rank- M collapse hands back the a_p spectrum, i.e. the symmetry-class support, not the rank. The floor here reads *symmetry*, not *number*. To recover rank we cannot push harder on this plane; we must add an axis.

Part 2 — add the axis: the same a_p , weighted, sees rank

The wall says rank is not in the distribution. So add the axis the distribution does not carry: the **analytic continuation** of $L(E, s)$, whose order of vanishing at $s = 1$ is the analytic rank (BSD: equal to the Mordell–

Weil rank). Its cheap, real-computable shadow is the **Mestre–Nagao sum** $S(X) = -\frac{1}{\log X} \sum_{p \leq X} \frac{a_p \log p}{p}$, which by the explicit formula tends to the rank (up to a known constant offset) as $X \rightarrow \infty$. This is the *same* a_p — now read through the p -weighting that connects the local data to the added complex axis.

```
# --- Part 2: Mestre-Nagao rank statistic on the SAME a_p, at growing budget ---
print("Nagao sum S(X) = -(1/log X) * sum_{p<=X} a_p log p / p\n")
budgets = [2000, 5000, 20000]
print(f" {'curve':8s}{'rank':6s}" + "".join(f"X={b:<10d}" for b in budgets))
nag = {}
for name, (xs, N, r) in apdata.items():
    row = {}
    for X in budgets:
        row[X] = sum(-ap*math.log(p)/p for (p, ap) in xs if p <= X)/math.log(X)
    nag[name] = row
    print(f" {'name':8s}{'r':<6d}" + "".join(f"{'row[b]:<12.2f}' for b in budgets))

Xmax = budgets[-1]
vals = [nag[n][Xmax] for n in ['11a1', '37a1', '389a1', '5077a1']]
monotone = all(vals[i] < vals[i+1] for i in range(3))
print(f"\nvalues at X={Xmax} by rank 0..3: {[round(v,2) for v in vals]}")
print(f"P2 (strictly monotone in rank): {'PASS' if monotone else 'FAIL'}")
```

Nagao sum $S(X) = -(1/\log X) * \sum_{p \leq X} a_p \log p / p$

curve	rank	X=2000	X=5000	X=20000
11a1	0	-0.10	-0.25	-0.22
37a1	1	0.82	0.80	0.65
389a1	2	1.89	1.92	1.83
5077a1	3	2.88	2.68	2.92

values at X=20000 by rank 0..3: [-0.22, 0.65, 1.83, 2.92]
P2 (strictly monotone in rank): PASS

Reading Part 2 — the wall dissolved

The ladder tracks the rank: $-0.2, 0.7, 1.8, 2.9$ for ranks $0, 1, 2, 3$ — strictly monotone, and close to the rank itself up to the known offset. What was *inexpressible* in the symmetry measure is an exact readout once the analytic axis is added. This is the "virtual memory" move on real arithmetic: the wall was not a failure, it was the coordinate surface where the real-line plane ran out of degrees of freedom, and the computation continues on a new axis.

The result is a **sorting**, and the sorting is the point:

reading of the same a_p	what it sees	what it is blind to	status
symmetry measure (moment clock)	Sato–Tate class (semicircle)	rank, conductor	universality LOCKED
arithmetic p -weighted sum (Nagao)	rank	fine symmetry detail	heuristic PULL

Rank does not live in the distribution of a_p ; it lives in the arithmetic weighting — the explicit-formula bridge to the L -function. The clock knowing only the symmetry class, and never the rank, is "shape before value" stated on arithmetic data: the geometry reads shape, the added analytic axis reads number.

Clay tie-in, and what stays gated

The same two-reading template is the pipeline for the Riemann gravity well. There, the *geometry* reading is the wall as a self-measuring counter, and the *arithmetic-weighted* reading is the prime-Stieltjes moment sequence whose wall tracks the zero-counting function $N(T)$ — the primes, weighted, reporting the count of zeros they cannot see directly. The BSD case in this notebook is the controlled, fully verifiable instance of that move: two readings of one local dataset, one blind to the fine invariant and one that recovers it by adding an axis.

The RH run itself — prime-Stieltjes moments, wall versus $N(T)$ — remains gated on two items before its numbers are trustworthy: the precision of the build-order run behind the earlier moment-accuracy figure (OP-1), and agreement on the moment convention between instances. This notebook does not attempt it; it establishes, on arithmetic where every claim is checkable, the reading structure that run will use — and it confirms the discipline that matters at the wall: the wall is where you add an axis, not where you stop.

